

Example Guide

ROS 2 For QArm

Getting Started:

Quanser provides a set of ROS 2 research examples for QArm. These examples demonstrate hardware interfaces and high-level control. All the packages provided for QArm are tested for **ROS 2 Kilted** and **Ubuntu 24.04**. To get started with ROS 2 please refer to ROS 2 documentation: <https://docs.ros.org/en/kilted>.

The QArm ROS 2 package is also dependent on Quanser Academic Resources python library, you can verify that it is added to the environment variable using the following command:

```
grep "Quanser" ~/.bashrc
```

If not, please add the following two lines to ~/.bashrc before running the ROS 2 examples:

```
export PYTHONPATH=$PYTHONPATH:/home/<your-  
username>/Documents/Quanser/0_libraries/python
```

```
export QAL_DIR=/home/<your-username>/Documents/Quanser
```

Package Location:

QArm ROS 2 nodes can be found as part of the available research examples provided by Quanser. Using file explorer, please navigate to the following resource directory:

~/Documents/Quanser/5_research/qarm/ROS Examples/ros2. The following packages are available for QArm:

- qarm_interfaces
- qarm_nodes

qarm_interfaces is a custom package for QArm-specific ROS 2 interface definitions. Custom messages include:

- **action/MoveQArm.action**: Encapsulates end-effector task-space motion goals and provides feedback and results during action execution.
- **msg/QArmDiagnostics.msg**: Contains low-level diagnostic information including joint currents, joint PWM values, and joint temperatures. It is intended for monitoring and safety supervision.

qarm_nodes package contains the runtime ROS 2 nodes and launch files that interface with QArm hardware and sensors:

- **qarm_hardware**: acts as the QArm hardware driver. It is responsible for receiving commands (joint angles, gripper, LED) and sending them to the QArm, and publishing sensor feedback including Joint States and QArm Diagnostics.

- **rgbd**: interfaces with the Intel RealSense RGBD camera mounted on the QArm. It publishes RGB images and depth images.
- **move_qarm_server**: this is the MoveQArm action server node. This node receives task space goal and sends joint commands to **qarm_hardware** node via ROS 2 topics. It also publishes action feedback and results during execution.
- **move_qarm_client**: this is an example MoveQArm action client node. It sends an action goal and monitors execution feedback and results.

Running Examples

Navigate to the QArm ROS 2 workspace (~/.Documents/Quanser/5_research/qarm/ROS Examples/ros2) and open a terminal session. Sourced **Kilted** as the desired ROS 2 distribution in your current terminal session.

```
source /opt/ros/kilted/setup.bash
```

If this is the first time running the QArm ROS packages, then compile the ros2 workspace.

```
colcon build --symlink-install
```

Once the workspace has compiled successfully it needs to be sourced as a ROS 2 package using the following command:

```
source install/setup.bash
```

To run any ROS 2 node, use the following command:

```
ros2 run <package name> <node name>
```

For example, to run the QArm hardware node, use the following command:

```
ros2 run qarm_nodes qarm_hardware
```

Note: you can only run one node per terminal session. If you want to run multiple nodes, either use multiple terminal sessions or combine them into launch files.

For example, after running the hardware node, if you want to run the Move Arm action server and client, open 2 addition terminal sessions, source the ROS 2 kilted directory and the QArm ROS packages for each terminal, and run the following:

```
ros2 run qarm_nodes move_qarm_server
```

```
ros2 run qarm_nodes move_qarm_client
```